

Student Number:

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Name:



2012

Preliminary Course

FINAL EXAMINATION

Tuesday, September 11

Mathematics

General Instructions

- Reading Time – 5 minutes.
- Working Time – 3 hours.
- Write using a black or blue pen.
- Approved calculators may be used.
- All necessary working should be shown for every question.

Total marks (100)

Section I

10 marks

- Attempt Questions 1 – 10
- Answer on the multiple choice answer sheet provided
- Allow approximately 15 minutes for this section

Section II

90 marks

- Attempt Questions 11 – 16
- Answer in the booklets provided
- Begin each question in a new booklet
- Allow approximately 2 hours 45 minutes for this section

- P2** provides reasoning to support conclusions which are appropriate to the context.
- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities.
- P4** chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques.
- P5** understands the concept of a function and the relationship between a function and its graph.

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for your responses to Questions 1 – 10.

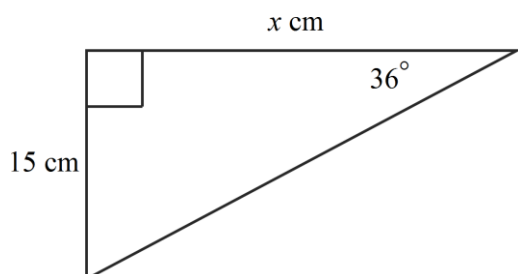
1. What is $\frac{3.23 \times 4.96^2}{3.45 + 1.2^2}$ correct to 3 significant figures?

(A) 16.2 (B) 16.3
(C) 24.4 (D) 24.5

2. Which of the following is the factorisation of $x^2 + x - 12$?

(A) $(x+3)(x-4)$ (B) $(x-3)(x-4)$
(C) $(x+6)(x-2)$ (D) $(x-3)(x+4)$

3. Which calculation would be used to find the value of x ?



(A) $15 \tan 36^\circ$ (B) $15 \cos 36^\circ$
(C) $\frac{15}{\tan 36^\circ}$ (D) $\frac{15}{\cos 36^\circ}$

4. What is the equation of the line that passes through the point (1, 3) and is parallel to the x -axis?

(A) $x = 1$ (B) $y = 1$ (C) $x = 3$ (D) $y = 3$

5. What are the values of a and b if $\frac{2\sqrt{2}}{2\sqrt{2} - 3} = a - \sqrt{b}$?

(A) -4 and 12 (B) -4 and 72 (C) -8 and 12 (D) -8 and 72

6. Which of the following are the solutions to the equation $2x^2 - 7x - 2 = 0$?

(A) $\frac{-7 \pm \sqrt{33}}{4}$

(B) $\frac{-7 \pm \sqrt{65}}{4}$

(C) $\frac{7 \pm \sqrt{33}}{4}$

(D) $\frac{7 \pm \sqrt{65}}{4}$

7. What is the domain for the function $y = \sqrt{7-x}$?

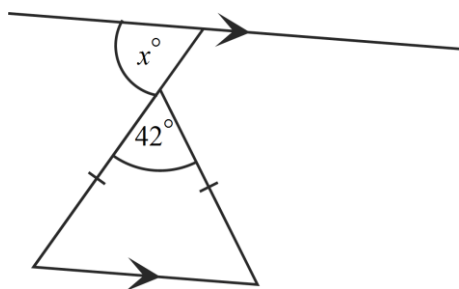
(A) $x \leq 7$

(B) $0 \leq x \leq 7$

(C) $x \geq 0$

(D) All real x

8.



What is the value of x in the above diagram?

(A) 42

(B) 69

(C) 111

(D) 138

9. For what values of k is the expression $\frac{-x^2}{4} - x - k$ negative definite?

(A) $k < 1$

(B) $k > 1$

(C) $k < 4$

(D) $k > 4$

10. What is $\frac{\sin(180^\circ - \theta)}{\sin(90^\circ - \theta)}$ in its simplest form?

(A) 1

(B) 2

(C) $\sin \theta$

(D) $\tan \theta$

End of Section I

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours 45 minutes for this section

Begin each question in a new booklet

All necessary working should be shown

All questions are of equal value

Question 11 (15 Marks)

Marks

- (a) Express $2.\dot{7}$ as a rational number in simplest form. 2

- (b) Solve for x :

(i) $\frac{5}{x+3} = \frac{3}{x}$ 2

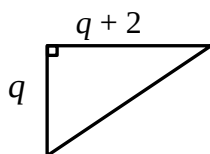
(ii) $|2x+1| = 3x+2$ 3

- (c) Show that the point $(1, 4)$ lies on the line $2x+3y-14=0$. 1

- (d) Show that the three points $A(3, -1)$, $B(5, 5)$ and $C(2, -4)$ are collinear. 2

- (e) If $\sin \theta = \frac{7}{25}$ and $\cos \theta < 0$ find the exact value of $\cot \theta$. 2

- (f) The diagram shows a right angled triangle in which one of the sides, adjacent to the right angle, is 2 centimetres longer than the other. 3



Find the length of the shortest side of this triangle if its area is 7.5 cm^2 .

End of Question 11

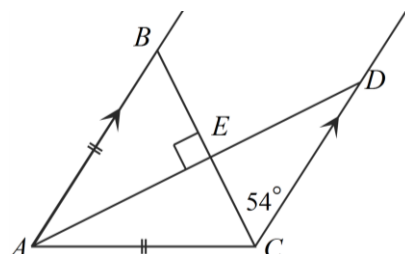
Question 12 (15 Marks) Start a new booklet**Marks**

- (a) An interval AB is formed by the points $A(-3, 5)$ and $B(2, -3)$ on the number plane.

Draw a clear diagram.

- (i) Find the gradient of the interval AB . 1
- (ii) Find the length of the interval AB . 1
- (iii) Show that the equation of the line AB is $8x + 5y - 1 = 0$. 1
- (iv) Find the equation of the line through $C(5, 8)$ perpendicular to AB . 2
- (v) Show that the perpendicular distance of C from the line AB is $\frac{79}{\sqrt{89}}$. 1
- (vi) Calculate the area of the triangle ABC . 1
- (vii) On your diagram, shade the region for which the inequalities $8x + 5y - 1 \geq 0$ and $y \leq 0$ hold simultaneously. 2
- (viii) The points A , B and C are three vertices of the parallelogram $ACBD$. Find the co-ordinates of D . 1
- (b) Simplify $\frac{1}{\operatorname{cosec}^2 \theta} + \frac{1}{\sec^2 \theta} + \frac{1}{\cot^2 \theta}$. 2

- (c) In the diagram, $AB \parallel CD$ and $AB = AC$.
The lines AD and BC are perpendicular and $\angle BCD = 54^\circ$.



Calculate, giving clear reasons, the size of $\angle CAD$.

3

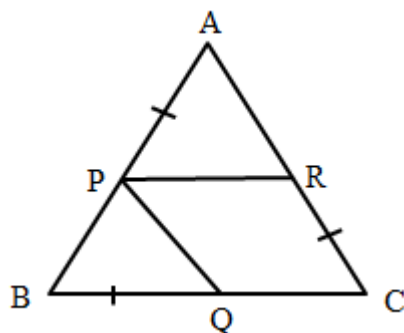
End of Question 12

Question 13 (15 Marks) Start a new booklet**Marks**

(a) Write down the exact value of $\tan 120^\circ$. 1

(b) Solve $4\cos^2\alpha = 3$ for the domain $0^\circ \leq \alpha \leq 360^\circ$. 3

(c) Triangle ABC is an equilateral triangle. The point P lies on the side AB , the point Q lies on the side BC and the point R lies on the side AC so that $AP = BQ = CR$.



NOT TO SCALE

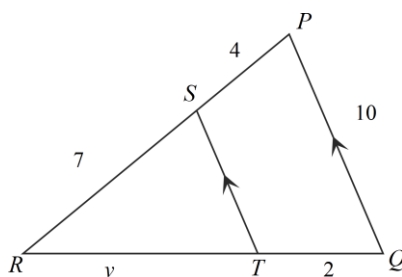
(i) Explain, with reasons, why $AR = BP$. 2

(ii) Prove $\triangle APR \equiv \triangle BQP$. 2

(d) (i) Solve simultaneously $y = 2x - 2$ and $y = x^2 - 1$. 2

(ii) Explain the graphical significance of your answer to part (i). 1

(e) In the triangle PQR , S and T are points on the sides PR and QR respectively such that $ST \parallel PQ$.



(i) Prove that the triangles PQR and STR are similar. 2

(ii) Find the value of y , giving reasons for your answer. 2

End of Question 13

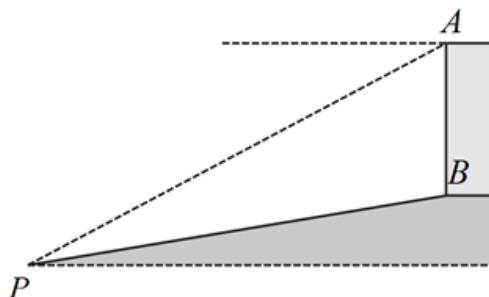
Question 14 (15 Marks) Start a new booklet**Marks**

- (a) The equation of a circle is given by $x^2 - 2x + y^2 + 12y - 12 = 0$.
- (i) Write the equation in the form $(x-h)^2 + (y-k)^2 = r^2$. 2
- (ii) Hence or otherwise write down the co-ordinates of the centre of the circle. 1
- (iii) Without solving, explain how you would determine the number of times a line cuts this circle. 2
- (b) Two ships leave a port, P , at the same time. Ship A travels at 12 km/h on a bearing of 213° . Ship B travels south-east at a speed of 8 km/h.
- (i) Draw a diagram to illustrate the relative positions of the two ships after 3 hours. 1
- (ii) Show that $\angle APB = 78^\circ$. 1
- (iii) Show that the distance between the ships after 3 hours is 39km, to the nearest km. 2
- (iv) Ship A received a distress call from Ship B whose engine had failed. On what bearing will Ship A need to sail in order to rescue Ship B ? Answer to the nearest degree. 3
- (c) The equations of two lines are $x + 2y - 6 = 0$ and $3x - 2y - 6 = 0$. 3
Find the equation of the line that passes through the point of intersection of these lines and the point $(1, -1)$.

End of Question 14

Question 15 (15 Marks) Start a new booklet**Marks**

- (a) A point P is at the foot of the hill which is inclined at 5° to the horizontal. The base B , of a tower AB , is situated 150 metres up the incline of the hill from P . From the top of the tower the angle of depression of the point P is 32° .



- (i) Copy the diagram and mark on it the size of $\angle PAB$ and $\angle APB$. 2
- (ii) Find the height of the tower, AB , correct to the nearest metre. 2
- (b) Consider the parabola with equation $y = x - x^2$.
- (i) Find the co-ordinates of the vertex. 2
- (ii) Sketch the parabola showing all important features. 2
- (c) A function is defined as
- $$f(x) = \begin{cases} x - 5 & x \geq 0 \\ -2 & -3 < x < 0 \\ 2 + x & x \leq -3 \end{cases}$$
- Find
- (i) $f(-1) + f(-5)$ 1
- (ii) Sketch $f(x)$ 3
- (d) Find the values of A , B and C such that 3
- $$3x^2 + 4x + 1 \equiv Ax(x - 2) + B(x + 1) + C$$

End of Question 15

Question 16 (15 Marks) Start a new booklet**Marks**

- (a) Solve the equation $4^x + 2^x - 6 = 0$. 3
- (b) If α and β are the roots of the equation $3x^2 + 5x - 4 = 0$, find
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\alpha^2 + \beta^2$ 2
- (iv) $\frac{3}{\alpha} + \frac{3}{\beta}$ 2
- (c) Given the equation $(k + 1)x^2 + (k + 2)x + k = 0$ find the value of k if the equation has equal roots. 3
- (d) The roots of the equation $x^2 - ax - 2b = 0$ differ by 4. 3
Show that $a^2 + 8b - 16 = 0$.

End of Examination Paper

Section I Answer Sheet

Student Number:

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use this multiple choice answer sheet for questions 1 – 10.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
correct

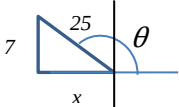
Completely fill the response oval representing the most correct answer.

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Solutions to Year 11 Mathematics 2012 Preliminary Examination

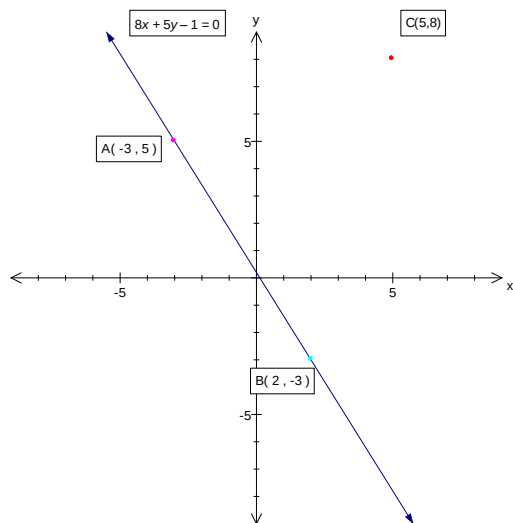
Multiple Choice	
1 B	2 D
3 C	4 D
5 D $\frac{2\sqrt{2}}{2\sqrt{2}-3} \times \frac{2\sqrt{2}+3}{2\sqrt{2}+3}$ $= \frac{2\sqrt{2}(2\sqrt{2}+3)}{(2\sqrt{2})^2 - (3)^2}$ $= \frac{8+6\sqrt{2}}{-1}$ $= -8 - \sqrt{72}$	6 B $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(-7) \pm \sqrt{(-7)^2 - 4(2)(-2)}}{2(-2)}$ $= \frac{7 \pm \sqrt{49+16}}{-4}$ $= \frac{7 \pm \sqrt{65}}{-4}$ $= \frac{-7 \pm \sqrt{65}}{4}$
7 A $7 - x \geq 0$ $x \leq 7$	8 B
9 A <i>For neg definite $b^2 - 4ac < 0$ and $a < 0$</i> $b^2 - 4ac < 0$ $(-1)^2 - 4\left(-\frac{1}{4}\right)(-k) < 0$ $1 - k < 0$ $k > 1$	
10 D $\frac{\sin(180 - \theta)}{\sin(90 - \theta)}$ $= \frac{\sin \theta}{\cos \theta}$ $= \tan \theta$	

Question 11	Marking Criteria
a) $\text{Let } x = 2.77777\ldots(1)$ $\text{then } 10x = 27.7777\ldots(2)$ $(2) - (1): 9x = 25$ $\therefore x = \frac{25}{9}$	2 - correct answer 1 - attempt at correct solution i.e correct x and $10x$
b) i) $\frac{5}{x+3} = \frac{3}{x}$ $5x = 3x + 9$ $2x = 9$ $x = \frac{9}{2}$	2 – correct solution 1 – one error in working
b) ii) $ 2x+1 = 3x+2$ $2x+1 = \pm(3x+2)$ $2x+1 = 3x+2 \text{ or } 2x+1 = -3x-2$ $-x = 1 \quad \text{or } 5x = -3$ $x = -1 \quad \text{or } x = \frac{-3}{5}$ But $3x+2 \geq 0$ $\text{i.e. } x \geq \frac{-2}{3}$ $\therefore x = \frac{-3}{5} \text{ is the only solution}$	3 – correct solution 2 – both values correct, no testing 1 – one value correct, no test
c) $\text{Let LHS} = 2x + 3y - 14$ $= 2(1) + 3(4) - 14$ $= 0$ $= \text{RHS}$ $\therefore (1, 4) \text{ lies on the line } 2x + 3y - 14 = 0$	1 – correct method or use of substitution

d) Compare the gradients of 2 intervals: m_{AB}, m_{BC} or m_{AC} <i>This solution looks at m_{AB} and m_{BC}</i> $m_{AB} = \frac{5-1}{5-3} \quad m_{BC} = \frac{5-4}{5-2}$ $= \frac{6}{2} \quad = \frac{9}{3}$ $= 3 \quad = 3$ $= \frac{6}{2}$ $= 3$ Since $m_{AB} = m_{BC}$, A, B and C are collinear	2 – correct answer showing 2 equal gradients 1 – incorrect but equal gradients or error in solution or other method with error
e) $\sin \theta = \frac{7}{25}$ implies θ in 1st & 2nd quadrant $\cos \theta < 0$ implies θ in 2nd & 3rd quadrant $\therefore \theta$ lies in 2nd quadrant $\therefore \cot \theta < 0$  By Pythagoras $25^2 = x^2 + 7^2$ $625 = x^2 + 49$ $x^2 = 576$ $x = 24$ $\therefore \cot \theta = -\frac{24}{7}$	2 – correct answer 1 – answer $+\frac{24}{7}$ or correct negative sign for 2nd quad
f) $A = \frac{1}{2}bh$ $\frac{q}{2}(q+2) = 7.5$ $q(q+2) = 15$ $q^2 + 2q - 15 = 0$ $(q-3)(q+5) = 0$ $\therefore q = 3, -5$ Since $q > 0$, the shortest side is 3cm.	3 – correct solution plus reason $q \neq -5$ 2 – correct values without consideration of the validity of the answer 1 – correct initial equation
Communication: 1 mark each (b) (ii) $3x + 2 \geq 0$ (c) clear reasoning eg LHS = ... = RHS (d) efficiency of soln eg 2 gradients only (e) stating 2nd quadrant (f) clear reasoning for one solution	

Question 12

(a)



(i)

$$m_{AB} = \frac{5 - (-3)}{-3 - 2}$$

$$= -\frac{8}{5}$$

1 – correct solution

(ii)

$$d_{AB} = \sqrt{(-3 - 2)^2 + (5 - (-3))^2}$$

$$= \sqrt{25 + 64}$$

$$= \sqrt{89}$$

1 – correct solution

(iii) **Method 1:**

$$m_{AB} = -\frac{8}{5} \text{ from (i)}$$

Equation AB:

$$y - y_1 = m(x - x_1)$$

$$y - 5 = -\frac{8}{5}(x + 3)$$

$$5y - 25 = -8x - 24$$

$$8x + 5y + 1 = 0$$

Method 2:

Equation AB is $8x + 5y - 1 = 0$

$$\text{Subs } A(-3, 5): LHS = 8(-3) + 5(5) - 1$$

$$= -24 + 25 - 1$$

$$= 0$$

$$= RHS$$

$$\text{Subs } B(2, -3): LHS = 8(2) + 5(-3) - 1$$

$$= 16 - 15 - 1$$

$$= 0$$

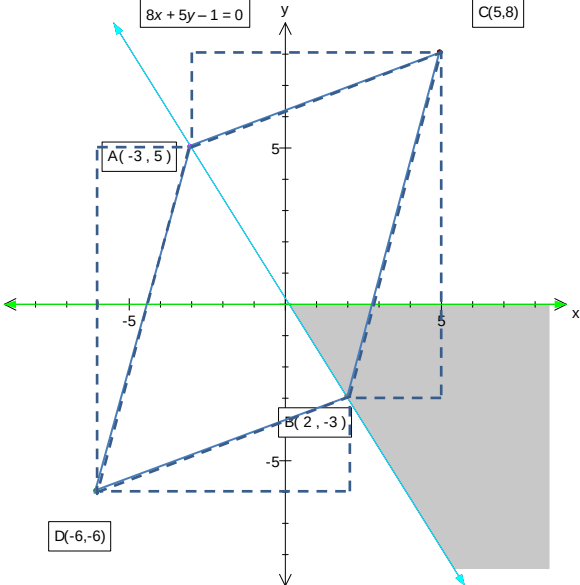
$$= RHS$$

$\therefore A$ and B satisfy $8x + 5y - 1 = 0$

$\therefore$ the equation AB is $8x + 5y - 1 = 0$

1 – correct solution

(iv) gradient line perpendicular to AB $m_1 \times m_2 = -1$ $m_{AB} = \frac{-8}{5}$ $\therefore$ gradient line perpendicular to $AB = \frac{5}{8}$ Equatⁿ required with $m = \frac{5}{8}$ and point $(5,8)$ $y - y_1 = m(x - x_1)$ $y - 8 = \frac{5}{8}(x - 5)$ $8y - 64 = 5x - 25$ $5x - 8y + 39 = 0$ OR Equatⁿ of a line perpendicular to $Ax + By + c = 0$ has form $Bx - Ay + k = 0$ $\therefore$ Req'd equation is $5x - 8y + k = 0$ subs $(5,8)$: $5(5) - 8(8) + k = 0$ $\therefore k = 39$ Req'd equation is $5x - 8y + 39 = 0$	2 – correct solution 1- correct with incorrect gradient
(v) $\text{Perp distance} = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 5(8) + 8(5) - 1 }{\sqrt{5^2 + 8^2}}$ $= \frac{ 40 + 40 - 1 }{\sqrt{25 + 64}}$ $= \frac{79}{\sqrt{89}}$	1 – correct substitution and answer
(vi) $\text{Area} = \frac{1}{2}bh$ $= \frac{1}{2}\sqrt{89} \times \frac{79}{\sqrt{89}}$ $= \frac{79}{2} \text{ units}^2$	1 – correct solution

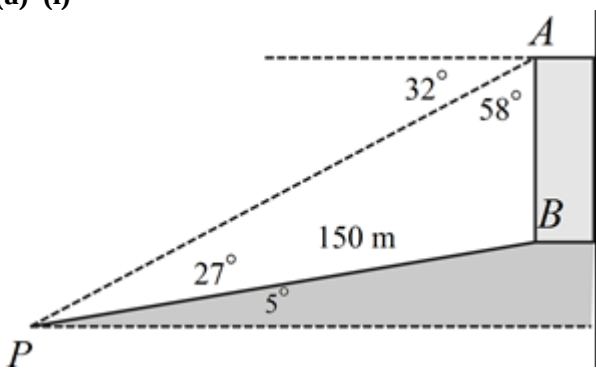
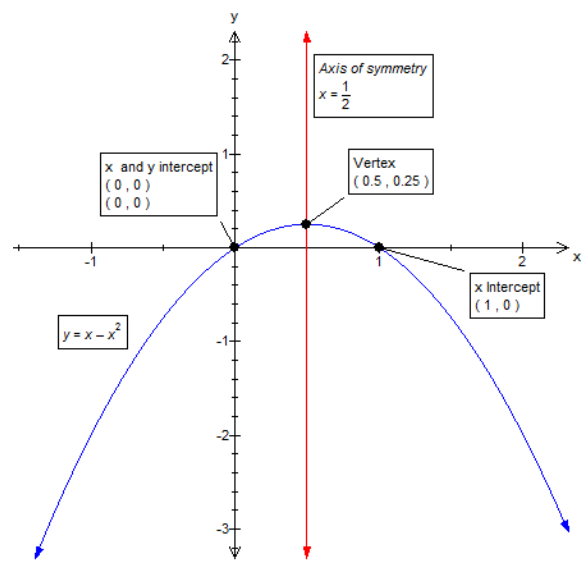
(vii) and (viii) 	vii) 2 – correct solution 1 – correct shading of one region viii) 1 – correct solution
(b) $\frac{1}{\operatorname{cosec}^2 \theta} + \frac{1}{\sec^2 \theta} + \frac{1}{\cot^2 \theta}$ $= \sin^2 \theta + \cos^2 \theta + \tan^2 \theta$ $= 1 + \tan^2 \theta$ $= \sec^2 \theta$	2 – correct substitution and simplification 1 – correct recognition of inverse ratios
(c) $\angle ABC = \angle DCB = 54^\circ$ (equal alternate angles, $AB \parallel CD$) $\angle ABC = \angle BCA = 54^\circ$ (angles opposite equal sides in isosceles $\triangle ABC$) * $\angle EAC = 36^\circ$ (exterior $\angle BEA$ equals the sum of the two interior angles) OR * $\angle EAC = 36^\circ$ (angle sum of $\triangle EAC$ is 180°)	3 – correct solution and clearly stated reasons 2 – one error in reasoning/calculation 1 – two errors in reasoning/calculation
Communication: 1 mark each (a) (iv) inclusion $m_1 m_2 = -1$ (vii) extension of interval AB to a line (viii) explanation of how students arrived at a solution (diagram, words, algebra) (b) efficient use of identities (c) efficient solution	

Question 13	
(a) $\tan 120^\circ = -\tan 60^\circ$ $= -\sqrt{3}$	1 – correct solution
(b) $4\cos^2 \alpha = 3$ $\cos \alpha = \pm \frac{\sqrt{3}}{2}$ $\therefore \alpha$ in all quadrants α in 1 st quadrant = 30° $\therefore \alpha = 30^\circ, 150^\circ, 210^\circ, 330^\circ$	3 – correct solution 2 - no $\pm$ mentioned but correct answers 1 – no $\pm$ mentioned incorrect related angle but angles correct using incorrect related angle
(c) (i) $AB = AC = BC$ (equal sides in equil $\triangle ABC$) $AP = RC = BQ$ (given) $AR = AC - RC$ $= AB - AP$ $= BP$ $\therefore AR = BP$	2 – correct solution with reasoning 1 – correct working with partial or no reasoning
(c) (ii) In $\triangle APR$ and $\triangle BQP$ $AP = BQ$ (given) $AR = BP$ (proven above) $\angle PAR = \angle QBP = 60^\circ$ (vertex angles of equilateral $\triangle ABC$) $\therefore \triangle APR \equiv \triangle BQP$ (SAS)	2 – correct proof with reasoning 1 – missing or incorrect reasons
(d) (i) $y = 2x - 2$(1) $y = x^2 - 1$(2) Subs (1) in (2) $x^2 - 1 = 2x - 2$ $x^2 - 2x + 1 = 0$ $(x - 1)^2 = 0$ $\therefore x = 1$	2 – correct solution 1 – most working correct or not finding the value of y .
(ii) The line is a tangent to the parabola	1 – correct answer
(e) (i) In $\triangle PQR$ and $\triangle STR$ $\angle R$ is common $\angle RPQ = \angle RST$ (equal corresponding angles on parallel lines, $PQ \parallel ST$) $\angle PQR = \angle STR$ (equal corresponding angles on parallel lines, $PQ \parallel ST$) $\therefore \triangle PQR \parallel \triangle STR$ (equiangular)	2 – correct proof with reasoning 1 – proof with most of the reasoning correct.

(ii) Since $\triangle PQR \parallel \triangle STR$, corr sides are in ratio $\frac{y}{y+2} = \frac{7}{11}$ $11y = 7y + 14$ $4y = 14$ $y = 3.5$ OR (ratio of intercepts made by parallel lines are equal) $\frac{y}{2} = \frac{7}{4}$ $y = 3.5$	2 – correct solution with reasoning 1 – correct answer with no reasoning
Communication: 1 mark each (b) diagram to show domain (c) (i) & (c) (ii) clear reasoning	(d) (ii) clear explanation (e) (i) efficiency of solution (e) (ii) efficiency of solution/ratio statement

Question 14	
(a) (i) $x^2 - 2x + y^2 + 12y - 12 = 0$ $(x^2 - 2x + 1) + (y^2 + 12y + 36) = 12 + 1 + 36$ $(x-1)^2 + (y+6)^2 = 49$	2- correct solution showing completing the square 1- progress towards completing the square
(ii) Centre = (1, -6)	1- correct solution
(iii) Find the perpendicular distance of the line from the centre of the circle and then compare that length with the radius of the circle (7) - if $d < 7$ the line cuts twice $d = 7$ line is a tangent to the circle $d > 7$ the line does not cut the circle Alternative solution would be to solve the equations of the line and the circle simultaneously. You would get a quadratic equation to solve. Using the discriminate we could calculate the number of roots. $\Delta < 0$ line does not cut circle $\Delta = 0$ line is a tangent to the circle $\Delta > 0$ line cuts the circle in 2 places	2 – correct explanation which includes how you got discriminant or perp dist of line to centre of circle 1 – explanation that was not clear
(b) (i) $A \text{ se} = 90^\circ + 45^\circ =$ $B \text{ } 213^\circ = 180^\circ + 33^\circ$	1 – diagram clearly indicating the length and bearing of each point

(ii) $\angle APB = 213^0 - 135^0$ $= 78^0$	1- correct show Alternate shows available
(iii) $AB^2 = 36^2 + 24^2 - 2(36)(24)\cos 78^0$ $= 1512.728598$ $AB = 38.8938..$ $= 39km \text{ (to nearest km)}$	2 – correct show which includes calculator readout 1 – correct substitution into cosine rule
(iv) $\frac{\sin A}{24} = \frac{\sin 78}{39}$ $\sin A = \frac{24 \sin 78}{39}$ $\angle A = 37^0 0' 31.5''$ $\text{Bearing} = 33^0 + 37^0$ $= 070^0$	3- correct 3 digit bearing 2- correct angle and attempt to use it to get bearing 1- correct attempt at sine rule
c) General form: $x + 2y - 6 + k(3x - 2y - 6) = 0$ Passes through (1, -1) $\therefore 1 + 2(-1) - 6 + k(3(1) - 2(-1) - 6) = 0$ $-7 - k = 0$ $k = -7$ $\therefore x + 2y - 6 - 7(3x - 2y - 6) = 0$ $x + 2y - 6 - 21x + 14y + 42 = 0$ $-20x + 16y + 36 = 0$ $5x - 4y - 9 = 0$ Or Point of intersection of lines (3, 1.5) Gradient 5/4	3- correct solution 2- correct attempt at solution 1- correct attempt at using K or finding point of intersection of the 2 lines
Communication: 1 mark each <u>except</u> 1(a) (iii)- 2 marks (a) (iii) clear reasoning (b) (i) diagram (b) (ii) clear reasoning	

Question 15	
(a) (i) 	2 – correct diagram 1 – 150 m in wrong position or angle APB labelled as 32°
(a) (ii) By the sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ $\frac{h}{\sin 27^\circ} = \frac{150}{\sin 58^\circ}$ $h = \frac{150 \sin 27^\circ}{\sin 58^\circ}$ $= 80.30....$ $= 80m \text{ (nearest metre)}$	2- correct solution, correct solution from incorrect diagram in part i) 1 – correct attempt with 1 error
(b) (i) Axis of symmetry OR by roots $x = \frac{-b}{a}$ $= \frac{1}{2}$ $y = \frac{1}{4}$ $\therefore \text{Vertex at } \left(\frac{1}{2}, \frac{1}{4}\right)$ solve: $x(x-1) = 0$ $x=0,1$ Vertex at $x = \frac{1}{2}$	2 – correct coordinates 1 – correct x coordinate, or correct values for x and y not written as coordinates.
(ii) 	2 – correct sketch showing all intercepts and coordinates of vertex 1 – incorrect scale or missing points, Diagram must be neat or only 1 mark given

(c) (i) $f(-1) + f(-5)$ $= -2 + (2 - 5)$ $= -5$	1 – correct answer
(c) (ii)	3 – correct sketch 2 – correct diagram with incorrect circles, or dotted line drawn for $y = -2$ 1 – correct lines ignoring domain
(d) $3x^2 + 4x + 1 \equiv Ax(x - 2) + B(x + 1) + C$ $= Ax^2 - 2Ax + Bx + B + C$ $= Ax^2 - x(2A + B) + B + C$ $\therefore A = 3 \quad -(2A + B) = 4 \quad B + C = 1$ $-6 - B = 4 \quad \therefore C = -9$ $\therefore B = 10$	3 – correct solution 2 – correct attempt at solution with one error 1 – correct expansion of quadratic leading to correct factorising using x
Communication: 1 mark each <u>except</u> 15(a) (i)- 2 marks (a) (i) all relevant information - 2 marks (b) (i) efficiency of solution (b) (ii) clarity of diagram (c) (ii) clarity of diagram	

Question 16	
(a) $4^x + 2^x - 6 = 0$ $(2^x)^2 + 2^x - 6 = 0$ Let $m = 2^x$ $m^2 + 2m - 6 = 0$ $(m + 3)(m - 2) = 0$ $m = -3, 2$ ie $2^x = -3, 2$ but $2^x > 0$ for all x $\therefore 2^x = 2$ $\therefore x = 1$	3 – correct solution including no solution for $2^x = -3$ 2 – two correct possible solutions 1 – correct values of m
(b) (i) $\alpha + \beta = \frac{-b}{a}$ $= \frac{-5}{3}$	1 – correct answer
(ii) $\alpha\beta = \frac{-4}{3}$	1 – correct answer
iii) $\alpha^2 + \beta^2$ $= (\alpha + \beta)^2 - 2\alpha\beta$ $= \frac{25}{9} - 2\left(\frac{-4}{3}\right)$ $= \frac{49}{9}$	2 – correct solution or carried errors from parts (i) or (ii) 1 – one error
(iv) $\frac{3}{\alpha} + \frac{3}{\beta}$ $= \frac{3\beta + 3\alpha}{\alpha\beta}$ $= \frac{3(\beta + \alpha)}{\alpha\beta}$ $= -5 \div \frac{-4}{3}$ $= \frac{15}{4}$	2 – correct solution or carried errors from parts (i) or (ii) 1 – one error

(c) $\Delta = b^2 - 4ac = 0$ $(k + 2)^2 - 4(k + 1)k = 0$ $k^2 + 4k + 4 - 4k^2 - 4k = 0$ $-3k^2 + 4 = 0$ $k^2 = \frac{4}{3}$ $k = \pm \frac{2}{\sqrt{3}}$	3 – correct solution and answers 2 – one correct solution or $3k^2 = 4$ 1 – correct discriminant = 0
(d) Let the roots be α and $\alpha + 4$ then sum : $\alpha + \alpha + 4 = a$ $2\alpha + 4 = a$ $\alpha = \frac{a - 4}{2}$ $\alpha = \frac{a}{2} - 2 \dots \dots (1)$ product : $\alpha(\alpha + 4) = -2b \dots \dots \dots (2)$ subs (1) in (2) : $\left(\frac{a}{2} - 2\right)\left(\frac{a}{2} - 2 + 4\right) = -2b$ $\left(\frac{a}{2} - 2\right)\left(\frac{a}{2} + 2\right) = -2b$ $\frac{a^2}{4} - 4 = -2b$ $a^2 - 16 = -8b$ $a^2 + 8b - 16 = 0$	3 – correct solution and answers 2 – correct substitution for α 1 – correct equations for sum and product
Communication: 1 mark each (a) appropriate substitution (b) (iii) use of identity (c) stated understanding of equal roots (d) efficiency of solution	